Low Power

Ideas

- Radix-4, 8 or16 – Franz-Josef,
- Clock Gating -
- Offload to Register – Jiahui, Ali ------ I think Romeu implemented this on github. Take a look (put code on gpt to explain)
- Trivial-Twiddle Fast Paths – Romeu---- works (RESULTS IN GITHUB)
- Remove recursive twiddle, and substitute for LUT – Romeu---- works (RESULTS IN GITHUB)
- Constant twiddle factors (Just wire connection for –1, 1, j, -j)
- Bit Width Reduction (Only if audio file is less than 32bit)
- Hardcoded Twiddles: since the FFT size is constant Replace the stage twiddle factors with fixed RTL constants instead of storing and reading them from accelerator memory. We can then reduce the SRAM cuz we will only need 64 words for the 32 complex values of the FFT.
- Since we are making the FFT of an audio signal, the output is symmetrical around the central frequency bin, so maybe we can just compute half the bins? (exploit symmetry)
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